METHOD 5: Using Python HTTPS server

1. In VirtualBox Manager open Settings → Network for each VM, set Adapter 1 → Attached to: Host-only Adapter and choose the host-only interface.
2. Start (or reboot) both VMs so the new host-only network takes effect.
3. Get the IP addresses
4. On VM1 install OpenSSL

```
└─$ sudo apt install openssl -y
openssl is already the newest version (3.2.2-1).
openssl set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

5. Create a folder to share and change to it

```
┌──(kali㉿kali)-[~]
└─$ mkdir ~/https_share

┌──(kali㉿kali)-[~]
└─$ cd ~/https_share
```

6. Generate a self-signed certificate (this produces server.pem with both key and cert)

```
┌──(kali㉿kali)-[~/https_share]
└─$ openssl req -new -x509 -keyout server.pem -out server.pem -days 365 -nod
es
```

7. Create a test file inside the share directory

```
┌──(kali㉿kali)-[~/https_share]
└─$ nano filet.txt
```

```
File  Actions  Edit  View  Help
  GNU nano 8.1
Hello
How are you
TCE
```

8. Create and run the HTTPS server on VM1

```
┌──(kali㉿kali)-[~/https_share]
└─$ nano https_server.py
```

```
import http.server,ssl
server_add=('0.0.0.0',8443)
httpd=http.server.HTTPServer(server_add,http.server.SimpleHTTPRequestHandler)
httpd.socket=ssl.wrap_socket(httpd.socket,certfile="server.pem",server_side=True)
print("Serving on https://0.0.0.0:8443")
httpd.serve_forever()
```

```
┌──(kali㉿kali)-[~/https_share]
└─$ python3 https_server.py
/home/kali/https_share/https_server.py:4: DeprecationWarning: ssl.wrap_socke
t() is deprecated, use SSLContext.wrap_socket()
  httpd.socket=ssl.wrap_socket(httpd.socket,certfile="server.pem",server_sid
e=True)
Serving on https://0.0.0.0:8443
192.168.56.103 - - [05/Sep/2025 09:13:25] "GET /filet.txt HTTP/1.1" 200 -
192.168.56.103 - - [05/Sep/2025 09:14:11] "GET /filet.txt HTTP/1.1" 200 -
```

9. On VM2 (client) — download the file from VM1

```
tce@tce-VirtualBox:~$ wget --no-check-certificate https://192.168.56.101:8443/f
ilet.txt
--2025-09-05 18:43:25--  https://192.168.56.101:8443/filet.txt
Connecting to 192.168.56.101:8443... connected.
WARNING: cannot verify 192.168.56.101's certificate, issued by ‘O=Internet Widg
its Pty Ltd,L=Madurai,ST=Tamilnadu,C=IN’:
  Self-signed certificate encountered.
    WARNING: certificate common name ‘’ doesn't match requested host name ‘192.
168.56.101’.
HTTP request sent, awaiting response... 200 OK
Length: 22 [text/plain]
Saving to: ‘filet.txt’

filet.txt           100%[===================>]      22  --.-KB/s    in 0s

2025-09-05 18:43:25 (1.49 MB/s) - ‘filet.txt’ saved [22/22]
```

```
tce@tce-VirtualBox:~$ cat filet.txt
Hello
How are you
TCE
```